COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/10/26	**DATE SUBMITTED:**	07/10/26
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ACTIVITY NO. HOA 1.1 Basic C++ Programming

1. PROCEDURE OUTPUT

Sections	Answers
Header File Declaration Section	`#include <iostream>` `#include <string>`
Global Declaration Section	

Class Declaration and Method Definition Section	`int sum(int fNum, int sNum) { return fNum + sNum; }`
	`std::string compare(int v1, int v2) { if (v1 > v2) { return "Variable A is greater than Variable B"; } else if (v1 < v2) { return "Variable B is greater than Variable A"; } else { return "Variable A and B are both equal"; } }`
	`bool compareBool(bool a, bool b) { std::cout << "a && b = " << (a && b) << std::endl; std::cout << "a || b = " << (a || b) << std::endl; std::cout << "!a = " << (!a) << std::endl; std::cout << "!b = " << (!b) << std::endl;`
	`return true; }`

Main Function

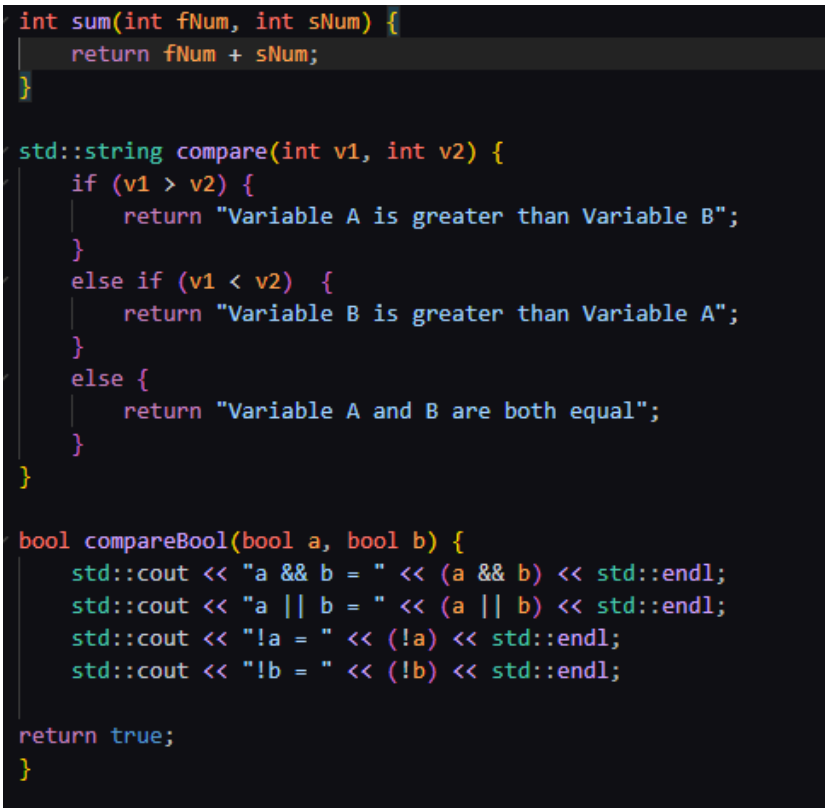
```
int main() {
    int fNum, sNum, v1, v2;
    bool a, b;
    std::cout << "Enter first number: ";
    std::cin >> fNum;
    std::cout << "Enter second number: ";
    std::cin >> sNum;
    std::cout << "The sum is: " << sum(fNum, sNum) << std::endl;

    std::cout << "Enter variable A: ";
    std::cin >> v1;
    std::cout << "Enter variable B: ";
    std::cin >> v2;
    std::cout << compare(v1, v2) << std::endl;

    std::cout << "Enter first boolean value(1 or 0): ";
    std::cin >> a;
    std::cout << "Enter second boolean value(1 or 0): ";
    std::cin >> b;
    std::cout << compareBool(a,b);

    return 0;


}
```

Method Definition	

```
int sum(int fNum, int sNum) {
    return fNum + sNum;
}

std::string compare(int v1, int v2) {
    if (v1 > v2) {
        return "Variable A is greater than Variable B";
    }
    else if (v1 < v2)  {
        return "Variable B is greater than Variable A";
    }
    else {
        return "Variable A and B are both equal";
    }
}

bool compareBool(bool a, bool b) {
    std::cout << "a && b = " << (a && b) << std::endl;
    std::cout << "a || b = " << (a || b) << std::endl;
    std::cout << "!a = " << (!a) << std::endl;
    std::cout << "!b = " << (!b) << std::endl;

return true;
}
```

Table 1-1. C++ Structure Code for Answer

ILO A: Create Code That Follows the Basic C++ Code Structure

- Create a function that will take two numbers and display the sum.

```
#include <iostream>
#include <string>

int sum(int fNum, int sNum) {
    return fNum + sNum;
}
```

- Create a function that will return whether variable A is greater than variable B.

```
std::string compare(int v1, int v2) {
    if (v1 > v2) {
        return "Variable A is greater than Variable B";
    }
    else  {
        return "Variable B is greater than Variable A";
    }
}
```

- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

```
bool compareBool(bool a, bool b) {
    std::cout << "a && b = " << (a && b) << std::endl;
    std::cout << "a || b = " << (a || b) << std::endl;
    std::cout << "!a = " << (!a) << std::endl;
    std::cout << "!b = " << (!b) << std::endl;

return true;
}
```

Screenshot of whole code and output:

Prelim_HOA 1.1 > hoa1.cpp > compareBool(bool, bool)

```cpp
#include <iostream>
#include <string>

int sum(int fNum, int sNum) {
    return fNum + sNum;
}

std::string compare(int v1, int v2) {
    if (v1 > v2) {
        return "Variable A is greater than Variable B";
    }
    else if (v1 < v2)  {
        return "Variable B is greater than Variable A";
    }
    else {
        return "Variable A and B are both equal";
    }
}

bool compareBool(bool a, bool b) {
    std::cout << "a && b = " << (a && b) << std::endl;
    std::cout << "a || b = " << (a || b) << std::endl;
    std::cout << "!a = " << (!a) << std::endl;
    std::cout << "!b = " << (!b) << std::endl;

return true;
}


```

```cpp
int main() {
    int fNum, sNum, v1, v2;
    bool a, b;
    std::cout << "Enter first number: ";
    std::cin >> fNum;
    std::cout << "Enter second number: ";
    std::cin >> sNum;
    std::cout << "The sum is: " << sum(fNum, sNum) << std::endl;

    std::cout << "Enter variable A: ";
    std::cin >> v1;
    std::cout << "Enter variable B: ";
    std::cin >> v2;
    std::cout << compare(v1, v2) << std::endl;

    std::cout << "Enter first boolean value(1 or 0): ";
    std::cin >> a;
    std::cout << "Enter second boolean value(1 or 0): ";
    std::cin >> b;
    std::cout << compareBool(a,b);

    return 0;


}
```

```
PS D:\GABRIEL FILES\A\CPE\2nd Year\1st Sem\CPE 010 - DSA> cd "d:\GABRIEL FI
E\2nd Year\1st Sem\CPE 010 - DSA\Prelim_HOA 1.1\" ; if ($?) { g++ hoa1.cpp
} ; if ($?) { .\hoa1 }
Enter first number: 1
Enter second number: 1
The sum is: 2
Enter variable A: 20
Enter variable B: 30
Variable B is greater than Variable A
Enter first boolean value(1 or 0): 1
Enter second boolean value(1 or 0): 0
a && b = 0
a || b = 1
!a = 0
!b = 1
1
PS D:\GABRIEL FILES\A\CPE\2nd Year\1st Sem\CPE 010 - DSA\Prelim_HOA 1.1>
```

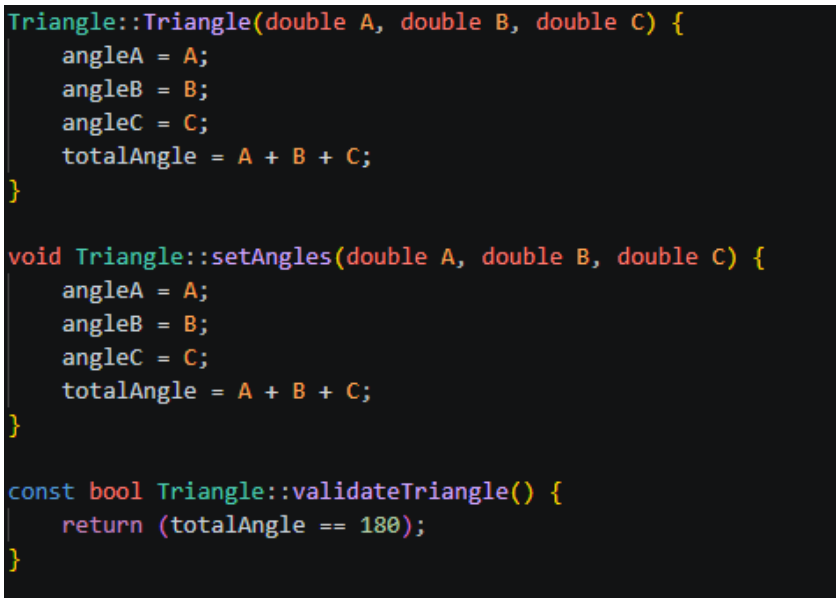
Code Analysis:
In the first task, I declared a function that adds the first number and second number. Whenever the user enter the first and second number, it automatically displays the sum. Next, it ask for the user to enter two variables (A & B) which then returns the value if a value is greater than the other or if they are equal. Lastly, the program asks the user again to enter a boolean value (1 or 0) then it returns 1 or 0 if it satisfies logical operations (AND, OR, NOT)

Table 1-2. ILO A output observations and comments.

Sections	Answers
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	`class Triangle {`
	`private:`
	`double totalAngle, angleA, angleB, angleC;`
	`public:`
	`Triangle(double A, double B, double C);`
	`void setAngles(double A, double B, double C);`
	`const bool validateTriangle();`
	`};`

Main Function	`int main(){`
	`//driver code`
	`Triangle set1(40, 30, 110);`
	`if (set1.validateTriangle()){`
	`std::cout << "The shape is a valid triangle " << std::endl;`
	`}`
	`else {`
	`std::cout << "The shape is NOT a valid triangle " << std::endl;`
	`}`
	`return 0;`
	`}`

Method Definition	

```
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}
```

Table 2-1. C++ Structure Code for Answer

Screenshot of whole code and output:

Prelim_HOA 1.1 > iloB.cpp > main()

```cpp
#include <iostream>

class Triangle {

private:
        double totalAngle, angleA, angleB, angleC;
public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};
Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}

int main(){

    Triangle set1(40, 30, 110);
    if (set1.validateTriangle()){
        std::cout << "The shape is a valid triangle " << std::endl;
    }
    else {
        std::cout << "The shape is NOT a valid triangle " << std::endl;
    }
    return 0;
}
```

```
The shape is a valid triangle
PS D:\GABRIEL FILES\A\CPE\2nd Year\1st Sem\CPE 010 - DSA\Prelim_HOA 1.1>
```

Code Analysis:

Based on what I've observed, a Triangle class was created with three angles and their total. The constructor sets the values of the angles when a triangle is made and it computes the sum if it is equal to 180. A setter function allows the angles to be changed later on. The code also has validation function which checks if the sum equals to 180 which can determine if the triangle is valid or not. In the main part of the program, a triangle with angles 40, 30, and 110 is tested, and the output shows if it is valid or not.

Table 1-2. ILO A output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve Different Problems using the C++ Programming Language

1. Create a C++ program to swap the two numbers in different variable

```cpp
#include <iostream>

int main(){
int a,b,c;
a = 8;
b = 7;
c = a;
a = b;
b = c;
std::cout << "a = " << a << std::endl;
std::cout << "b = " << b << std::endl;
}
```

```
PS D:\GABRIEL FILES\A\C
\CPE\2nd Year\1st Sem\C
?) { .\swap }
a = 10
b = 5
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit

Prelim_HOA 1.1 > temperature.cpp > main()

```cpp
#include <iostream>

double convertKelvinToFahrenheit(double K) {
    return (K - 273.15) * 9.0 / 5.0 + 32.0;
}

int main() {
    double K;
    std::cout << "Enter temperature in Kelvin: ";
    std::cin >> K;
    std::cout << "Temperature in Fahrenheit: " << convertKelvinToFahrenheit(K)  << std::endl;
}
```

```
ture } ; if ($?) { .\temperature }
Enter temperature in Kelvin: 0
Temperature in Fahrenheit: -459.67
```

3. Create a C++ program that has a function that will calculate the distance between two points.

Prelim_HOA 1.1 > distance.cpp > ...

```cpp
#include <iostream>
#include <cmath>

float calculateDistance(float x1, float y1, float x2, float y2) {
    return sqrt(pow(x2 - x1, 2) + pow(y2 - y1,2));
}

int main() {
    float x1, y1, x2, y2;
    std::cout << "Enter the coordinates of the first point (x1, y1): ";
    std::cin >> x1 >> y1;
    std::cout << "Enter the coordinates of the second point (x2, y2): ";
    std::cin >> x2 >> y2;
    std::cout << "The distance between the two points is: " << calculateDistance(x1, y1, x2, y2) << std::endl;
    return 0;
}
```

```
Enter the coordinates of the first point (x1, y1): 1 2
Enter the coordinates of the second point (x2, y2): 3 4
The distance between the two points is: 2.82843
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
double calculateArea(double base, double height) {
    return (0.5 * base * height);
}

double calculatePerimeter(double sideA, double sideB, double sideC) {
    return (sideA + sideB + sideC);
}

std::string classifyTriangleAngles(double angleA, double angleB, double angleC
    if (angleA < 90 && angleB < 90 && angleC < 90) {
        return "The triangle is an acute triangle.";
    } else if (angleA == 90 || angleB == 90 || angleC == 90) {
        return "The triangle is a right triangle.";
    } else {
        return "The triangle is an obtuse triangle.";
    }
}
```

```
Enter the angles of the triangle: 40 30 110
The shape is a valid triangle
Enter base and height of the triangle: 10 20
The area of the triangle is: 100
Enter the lengths of the three sides of the triangle: 10 20 30
The perimeter of the triangle is: 60
The triangle is an obtuse triangle.
```

Conclusion

In this hands-on activity, I was able to recall several lessons from when we first started learning the C++ programming language. I realized that input handling can be simplified — instead of asking for values one by one, such as the base and height of a triangle, you can enter two values separated by a space in a single prompt. I also reinforced my understanding of functions, which make coding more efficient since they allow me to reuse logic without rewriting it from scratch. For example, in ILO A, we created different functions for tasks such as computing sums, comparing variables, and handling Boolean operations.The procedure guided me through progressively challenging tasks, starting with simpler problems and moving toward more complex ones. The supplementary activity further allowed me to gauge my ability to apply C++ concepts in practice. At first, I struggled to recall certain formulas and logic, but with persistence I was able to remember them. The tasks became progressively more demanding, with Task 4 being the most challenging since I was unsure which functions or logic to use to make the code run correctly. I sought guidance from AI but intentionally avoided asking for the complete solution, as I wanted to learn by applying the hints myself. Overall, I believe I did fairly well in this activity. I managed to recall key concepts, apply them in practice, and overcome difficulties

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: ChatGPT, Microsoft Copilot, Quillbot

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Grammar correction, code assistance, explanation of concepts

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: I used ChatGPT, Microsoft Copilot, and QuillBot while working on this activity. These tools were really helpful in different ways. ChatGPT and Copilot gave me explanations and guided me through coding logic when I got stuck, while QuillBot helped me polish my grammar and sentence flow so my reflection would sound clearer.

My main purpose in using these tools was for grammar correction, idea generation, and

code assistance. Sometimes I needed help recalling formulas or understanding how to structure functions in C++, and the AI gave me hints without handing me the full solution. I also used them for summarization and concept explanation, which made it easier to organize my thoughts when writing reflections.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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